Spectral Energy Landscapes, Multiscale Window Entropies, and Continuous Bubble Relaxation of Magic Squares

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Abstract

Every magic square of order n is a zero-energy ground state of the line-indicator Hamiltonian $\mathcal{H}_{\text{line}}(S) = \sum_{L\in\mathcal{L}_n} R_L(S)^2 = \mathbf{z}^\top L_n \mathbf{z}$ on the symmetric group S_{n^2}, where $\mathbf{z} \in \mathbb{R}^{n^2}$ is the centred height field $Z_{ij} = 2s_{ij} - (n^2+1)$ and $L_n = A_n^\top A_n$ is the $(2n+2)$-line incidence Laplacian. While previous papers in this series established the convex-polytopal geometry of Magic Gems (Paper I), their universal bivariate polynomial moments and isotropic inertia tensors (Paper II), and their rigid-body Euler–Poinsot rotational mechanics (Paper III), a fundamental inverse question remained open: how does the combinatorial energy landscape $\mathcal{H}_{\text{line}} : S_{n^2} \to \mathbb{Z}_{\geq 0}$ organize the $n^2!$ permutations around the magic ground-state manifold, and how can physical relaxation dynamics navigate its frustrated local minima? Here we resolve this question across four scales. First, we prove the *Spectral Rationalization Theorem*: although L_n on $\mathbb{R}^{n^2}$ possesses irrational eigenvalues $\lambda_\pm(n)$ arising from row–column–diagonal overlap, restricting L_n to the (n^2-1)-dimensional zero-mean hyperplane $\mathbf{1}^\perp$ inhabited by all centred permutations decouples the constant mode and collapses the entire spectrum into exact integers—$\{0^{[n^2-2n-1]}, (n-2)^{[1]}, n^{[2n-1]}\}$ for even $n \geq 4$ and $\{0^{[n^2-2n-1]}, (n-1)^{[2]}, n^{[2n-2]}\}$ for odd $n \geq 3$—with a universal mean eigenvalue $\bar{\mu}(\widetilde{L}_n) = 2$ and exact random-permutation ensemble expectation $\mathbb{E}[\mathcal{H}_{\text{line}}(S)] = 2\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4-1)$. We also prove a *Low-Mode Insufficiency Proposition*, constructing an explicit 4×4 counterexample S_{low} whose dipole moments $\langle X, Z\rangle = \langle Y, Z\rangle = 0$ and main diagonals $R_{D_\pm} = 0$ vanish identically, yet which remains trapped at $\mathcal{H}_{\text{line}}(S_{\text{low}}) = 1{,}584 > 0$ inside the $(2n-3)$-dimensional high-frequency residual subspace. Second, we introduce multiscale sliding-window energies and Shannon entropies $H_p(k)$ across all 880 Frénicle order-4 magic squares and constructive families at $n \in \{3,4,5,6,8\}$, proving that vanishing interior 2×2 quartet energy ($W^{\text{int}}_{2\times 2} = 0$, 9 windows) is equivalent on magic squares to vanishing toroidal quartet energy ($W^{\text{tor}}_{2\times 2} = 0$, 16 windows) and holds on exactly the 48 panmagic squares of Dudeney Group I, while the mixed octupole moment M_{xyz} vanishes on all 48 Group III associative squares plus exactly 8 anomalous non-associative squares (4 each in Groups IX and X). Third, we derive the exact quadratic transposition law $\Delta\mathcal{H}_{\text{line}}(\tau_{ab}) = -2d(g_a - g_b) + c_{ab}d^2$ and perform an exhaustive census of all $9! = 362{,}880$ states of S_9, discovering that $n = 3$ possesses *exactly* 8 *spurious local minima* forming a single dihedral orbit $S^{(3)}_{\text{trap}}$ at $\mathcal{H}_{\text{line}} = 8$ that traps 25.47% of S_9 under both 2-cycle and 3-cycle descent and requires a simultaneous $(2,2)$ double transposition to reach the Lo Shu ground state. Fourth, we formulate a continuous "Bubble Trinity" Langevin relaxation dynamics across four regimes and demonstrate that hybrid field-guided $2+3$-cycle settling boosts order-4 ground-state synthesis from 48.7% to 69.0%.

1 Introduction

A magic square of order n is a bijection $S : \{1,\dots,n\}^2 \to \{1,\dots,n^2\}$ whose n rows, n columns, and two main diagonals each sum to the magic constant $M_n = \frac{1}{2}n(n^2+1)$ [7, 4, 6, 1]. When lifted into three-dimensional Euclidean space via centred integer coordinates

$$X_j = 2j - (n+1), \qquad Y_i = (n+1) - 2i, \qquad Z_{ij} = 2s_{ij} - (n^2+1), \tag{1}$$

each permutation $S \in S_{n^2}$ defines an n^2-point configuration $\mathcal{P}(S) = \{(X_j, Y_i, Z_{ij})\} \subset \mathbb{Z}^3$ whose convex hull $K(S) = \operatorname{conv} \mathcal{P}(S)$ is a *Magic Gem*. Across the first three papers of this series, we investigated the forward map from magic squares to geometric and mechanical invariants:

1. **Paper I (*Polytopal Geometry*)** [11] classified the facial structure, vertex counts V, surface areas, volumes, and discrete symmetry groups of Magic Gems across the 880 Frénicle order-4 squares and higher-order families.

2. **Paper II (*Bivariate Moments & Orthogonality*)** [10] proved that semi-magic balance forces $\langle X^j, Z\rangle = \langle Y^j, Z\rangle = 0$ for all $j \geq 0$, diagonalizing the 3×3 inertia and covariance tensors of every magic square and generating universal Prouhet–Tarry–Escott power-sum identities.

3. **Paper III (*Rigid-Body Mechanics & Structural Stability*)** [9] tested these algebraic invariants in physical rigid-body mechanics, proving that every equal-vertex-mass Magic Gem is an oblate symmetric top ($I_{xx} = I_{yy} < I_{zz}$) free of intermediate-axis Dzhanibekov tumbling, while continuous bulk polyhedra break covariance isotropy yet preserve exact static stability across all 50,334 facets of the 880 order-4 gems.

Despite this complete characterization of the *ground states*, a central statistical-mechanical and algorithmic question remained open: *how are the magic ground states embedded within the ambient $n^2!$-state permutation space S_{n^2}, and what physical or combinatorial dynamics can relax an arbitrary disordered permutation into a magic square?* Treating magic square synthesis as energy minimization on a frustrated discrete lattice connects classical combinatorics to spin-glass theory [13], rugged potential energy landscapes [17], and stochastic annealing [8]. In an interactive companion laboratory (`physicalSolver` [12]), we prototyped four complementary physical representations of this inverse problem: a spectral operator decomposition of line residuals, a multiscale sliding-window entropy analyzer, a discrete permutation-cycle descent engine, and a continuous "Bubble Trinity" Langevin particle simulator.

Following Whitesides's blueprint-driven methodology for connecting mathematical structure to quantitative experiments [18], this fourth paper formalizes the mathematical operators underlying `physicalSolver` and establishes exact theorems and exhaustive computational censuses for each stage of the relaxation hierarchy. Let $\mathcal{L}_n$ denote the set of $2n+2$ magic lines (n rows, n columns, and 2 main diagonals), and let $R_L(S) = \sum_{(i,j)\in L} Z_{ij}$ be the centred residual of line $L \in \mathcal{L}_n$. Collecting the $2n+2$ line indicator vectors as rows of the incidence matrix $A_n \in \{0,1\}^{(2n+2)\times n^2}$, the *line-indicator Hamiltonian* takes the quadratic form

$$\mathcal{H}_{\text{line}}(S) = \sum_{L\in\mathcal{L}_n} R_L(S)^2 = \|A_n \mathbf{z}\|_2^2 = \mathbf{z}^\top L_n \mathbf{z}, \qquad L_n = A_n^\top A_n \in \mathbb{Z}^{n^2\times n^2}. \tag{2}$$

By construction, $\mathcal{H}_{\text{line}}(S) \geq 0$ for all $S \in S_{n^2}$, and $\mathcal{H}_{\text{line}}(S) = 0$ if and only if S is a magic square.

Summary of Main Results

Our investigation proceeds from global linear operators on $\mathbb{R}^{n^2}$ to local multiscale sub-windows, discrete swap-graph topology on S_{n^2}, and continuous Langevin relaxation (Table 1):

1. **Spectral Rationalization & Low-Mode Insufficiency (Section 2):** In Theorem 2.1, we prove that while the unprojected Laplacian L_n on $\mathbb{R}^{n^2}$ has irrational eigenvalues $\lambda_\pm(n)$ caused by the coupling between the constant vector $\mathbf{1}$ and the two main diagonals, restricting L_n to the zero-mean hyperplane $\mathbf{1}^\perp$ via $\widetilde{L}_n = P_{\mathbf{1}^\perp} L_n P_{\mathbf{1}^\perp}$ collapses all eigenvalues into exact integers: $\{0^{[n^2-2n-1]}, (n-2)^{[1]}, n^{[2n-1]}\}$ for even $n \geq 4$ and $\{0^{[n^2-2n-1]}, (n-1)^{[2]}, n^{[2n-2]}\}$ for odd $n \geq 3$. In Theorem 2.2, we show that $\operatorname{tr}(\widetilde{L}_n) = 2(n^2-1)$ for all $n \geq 3$, so the mean eigenvalue of $\widetilde{L}_n$ on $\mathbf{1}^\perp$ is *identically* 2, yielding the exact random-permutation expectation $\mathbb{E}[\mathcal{H}_{\text{line}}(S)] = 2\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4-1)$ (Figure 1a). In Proposition 2.3, we prove that enforcing

Table 1: Overview of the four-scale physical solver hierarchy and principal theorems established in Paper D.

Scale / Section	Operator / Observable	Exact Theorem / Census	Key Quantitative Invariant
Sec. 2: Spectral Operator on $\mathbf{1}^\perp$	Line Laplacian $L_n = A_n^\top A_n$ Random ensemble $\mathbb{E}[\mathcal{H}_{\text{line}}]$ Dipole + diag energy E_{low}	Theorem 2.1 ($\mathbf{1}^\perp$ rationalization) Theorem 2.2 (universal mean $\bar{\mu} = 2$) Proposition 2.3 (low-mode insufficiency)	$\text{spec}(\widetilde{L}_n) \subset \mathbb{Z}$, $\ker = n^2 - 2n - 1$ $\mathbb{E}[\mathcal{H}_{\text{line}}] = 2\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4 - 1)$ S_{low}: $E_{\text{low}} = 0$, $\mathcal{H}_{\text{line}} = 1{,}584$
Sec. 3: Multiscale Windows & Parity	2×2 quartet energy $W_{2\times 2}$ Complement breaks B_{comp}, M_{xyz} Window entropy $\widetilde{H}_p(k)$	Theorem 3.1 (interior $\Leftrightarrow$ toroidal) Proposition 3.2 (octupole zero census) Constructive census ($n = 3 \ldots 8$)	$W_{2\times 2}^{\text{int}} = 0 \iff$ Group I (48/880) $M_{xyz} = 0$ on 48 Grp III +8 Grp IX/X Max $\widetilde{H}_1(2) = 0.99967$ (`#123`, Grp VI)
Sec. 4: Swap Landscape on S_{n^2}	Transposition cost $\Delta\mathcal{H}_{\text{line}}(\tau_{ab})$ S_9 exhaustive basin census S_{16} descent (2,000 starts)	Proposition 4.1 (exact quadratic law) Theorem 4.2 (8 antipodal traps $S_{\text{trap}}^{(3)}$) Order-4 local-minimum spectrum	$\Delta\mathcal{H}_{\text{line}} = -2d(g_a - g_b) + c_{ab}d^2$ 74.53% Lo Shu vs. 25.47% trapped 18.25% ground state; 81.75% trapped
Sec. 5: Continuous Bubble Relaxation	Bubble-Trinity Langevin SDE Field-guided $2+3$-cycle settle	Regimes I–IV Solver benchmark ($B = 200 \ldots 5{,}000$)	$\mathcal{H}_{\text{ind}}$: 1681.9 $\to$ 1050.6 (Regime IV) 48.7% $\to$ 69.0% recovery at $B = 5{,}000$

Paper II's dipole orthogonality $\langle X, Z\rangle = \langle Y, Z\rangle = 0$ together with diagonal balance $R_{D_+} = R_{D_-} = 0$ constrains only a 4-dimensional subspace of the $(2n+1)$-dimensional active image of $\widetilde{L}_n$, leaving a $(2n-3)$-dimensional high-frequency residual subspace exemplified at $n = 4$ by an explicit permutation S_{low} with $\mathcal{H}_{\text{line}}(S_{\text{low}}) = 1{,}584$ (Figure 1b).

2. **Multiscale Window Entropies, Quartet Equivalence & Octupole Census (Section 3):** Moving beyond 1D line sums to 2D local patches, we analyze sliding $k \times k$ sub-window energies and normalized Shannon entropies $\widetilde{H}_p(k)$ across constructive families ($n = 3, 4, 5, 6, 8$) and all 880 order-4 squares (Figure 2). In Theorem 3.1, we prove that for any 4×4 magic square, vanishing of the 9 interior 2×2 quartet sums ($W_{2\times 2}^{\text{int}}(S) = 0$) automatically forces all 7 boundary-wrapping toroidal 2×2 quartets to vanish ($W_{2\times 2}^{\text{tor}}(S) = 0$), isolating the 48 panmagic squares of Dudeney Group I. In Proposition 3.2, we show that the centrosymmetric complement-break count $B_{\text{comp}}(S) \in \{0, 8\}$ is binary across all 880 squares and that the mixed octupole moment $M_{xyz}(S) = \sum_{i,j} X_j Y_i Z_{ij}$ vanishes on all 48 Group III associative squares plus exactly 8 non-associative squares in Groups IX and X (Figure 3).

3. **Combinatorial Swap Landscape & The 8 Antipodal Traps of S_9 (Section 4):** In Proposition 4.1, we derive the exact closed-form energy change $\Delta\mathcal{H}_{\text{line}}(\tau_{ab}) = -2d(g_a - g_b) + c_{ab}d^2$ under any 2-cycle transposition τ_{ab}. Exhaustive evaluation across all $9! = 362{,}880$ permutations of S_9 (Theorem 4.2 and Figure 4) reveals a striking topological obstruction: besides the 8 Lo Shu ground states ($\mathcal{H}_{\text{line}} = 0$), S_9 contains *exactly* 8 *spurious strict local minima* forming a single D_4 orbit $S_{\text{trap}}^{(3)}$ at $\mathcal{H}_{\text{line}} = 8$ ($E_{\text{std}} = 2$). Every single 2-cycle AND every single 3-cycle on $S_{\text{trap}}^{(3)}$ strictly increases $\mathcal{H}_{\text{line}}$; escaping to Lo Shu requires a simultaneous $(2, 2)$ double swap. Under steepest descent, 74.53% (270,442) of S_9 flows to Lo Shu while 25.47% (92,438) is captured by $S_{\text{trap}}^{(3)}$. At order $n = 4$ (S_{16}), pure 2-cycle steepest descent reaches $\mathcal{H}_{\text{line}} = 0$ on only 18.25% of random starts (Figure 5).

4. **Continuous "Bubble Trinity" Langevin Relaxation & Hybrid Settling (Section 5):** Finally, we embed the discrete grid into $\mathbb{R}^2$ by treating the n^2 integers as mobile viscoelastic "bubbles" coupled to home lattice wells, pairwise core repulsion, and a continuous line-residual force field (Figure 6). Across four dynamical regimes (*Frozen Lattice*, *Slow Drift*, *Field Storm*, and *Centrifugal Balance*), we show how continuous stress-field pre-sorting combined with discrete $2+3$-cycle settling increases the order-4 ground-state recovery rate from 48.7% to 69.0% at $B = 5{,}000$ steps (and by 3.5× at $B = 500$ steps).

2 Spectral Rationalization of the Line-Incidence Laplacian

To analyze the global geometry of the line-indicator Hamiltonian $\mathcal{H}_{\text{line}}(S) = \mathbf{z}^\top L_n \mathbf{z}$ defined in Equation 2, we first determine the exact spectral decomposition of the line-incidence Laplacian $L_n = A_n^\top A_n$ both on the ambient Euclidean space $\mathbb{R}^{n^2}$ and on the zero-mean hyperplane $\mathbf{1}^\perp = \{\mathbf{z} \in \mathbb{R}^{n^2} : \mathbf{1}^\top \mathbf{z} = 0\}$ where every centred permutation vector $\mathbf{z}(S)$ resides.

2.1 Incidence Structure and Spectral Rationalization on $\mathbf{1}^\perp$

Let $\mathbf{r}_1, \ldots, \mathbf{r}_n \in \{0,1\}^{n^2}$ denote the row-indicator vectors of the $n \times n$ grid, $\mathbf{c}_1, \ldots, \mathbf{c}_n \in \{0,1\}^{n^2}$ the column-indicator vectors, and $\mathbf{d}_+, \mathbf{d}_- \in \{0,1\}^{n^2}$ the main and anti-diagonal indicator vectors. Then

$$L_n = A_n^\top A_n = \sum_{i=1}^{n} \mathbf{r}_i \mathbf{r}_i^\top + \sum_{j=1}^{n} \mathbf{c}_j \mathbf{c}_j^\top + \mathbf{d}_+ \mathbf{d}_+^\top + \mathbf{d}_- \mathbf{d}_-^\top. \tag{3}$$

Because the n rows partition the n^2 cells and the n columns also partition the n^2 cells, $\sum_{i=1}^{n} \mathbf{r}_i = \sum_{j=1}^{n} \mathbf{c}_j = \mathbf{1}_{n^2}$. This single linear dependency reduces the rank of the $2n$ row and column indicators from $2n$ to $2n-1$. Since the two main diagonals $\mathbf{d}_+$ and $\mathbf{d}_-$ are linearly independent of the row and column span for all $n \geq 3$, $\operatorname{rank}(A_n) = 2n+1$. However, because $\mathbf{d}_+$ and $\mathbf{d}_-$ have non-zero inner product $\mathbf{1}^\top \mathbf{d}_\pm = n$ with the constant vector $\mathbf{1}$ (and intersect at the central cell when n is odd, $\mathbf{d}_+^\top \mathbf{d}_- = 1$, or are disjoint when n is even, $\mathbf{d}_+^\top \mathbf{d}_- = 0$), the unprojected operator L_n on $\mathbb{R}^{n^2}$ mixes the constant mode $\mathbf{1}$ with the symmetric diagonal mode $\mathbf{d}_+ + \mathbf{d}_-$, producing irrational eigenvalues. Restricting to $\mathbf{1}^\perp$ eliminates this coupling completely:

Theorem 2.1 (Spectral Rationalization of the Line-Incidence Laplacian)**.** *For every order $n \geq 3$, the line-incidence matrix $A_n \in \{0,1\}^{(2n+2)\times n^2}$ has rank $2n+1$, and the magic nullspace $\ker(L_n) = \ker(A_n)$ has dimension*

$$\dim \ker(L_n) = n^2 - 2n - 1, \tag{4}$$

all of whose basis vectors lie in the zero-mean hyperplane $\mathbf{1}^\perp$. Furthermore:

1. ***Unprojected Spectrum on $\mathbb{R}^{n^2}$:*** *The $2n+1$ non-zero eigenvalues of L_n on $\mathbb{R}^{n^2}$ consist of two irrational roots $\lambda_\pm(n)$ together with integer eigenvalues:*

$$\operatorname{spec}(L_n) = \begin{cases} \left\{0^{[n^2-2n-1]},\ \lambda_-(n)^{[1]},\ n^{[2n-1]},\ \lambda_+(n)^{[1]}\right\}, & \text{even } n \geq 4, \\ \left\{0^{[n^2-2n-1]},\ \lambda_-(n)^{[1]},\ (n-1)^{[1]},\ n^{[2n-2]},\ \lambda_+(n)^{[1]}\right\}, & \text{odd } n \geq 3, \end{cases} \tag{5}$$

   where $\lambda_\pm(n) = \frac{1}{2}\big(3n \pm \sqrt{n(n+16)}\big)$ for even n and $\lambda_\pm(n) = \frac{1}{2}\big(3n+1 \pm \sqrt{n^2+14n+1}\big)$ for odd n.

2. ***Rationalized Zero-Mean Spectrum on $\mathbf{1}^\perp$:*** *Let $P_{\mathbf{1}^\perp} = I_{n^2} - \frac{1}{n^2}\mathbf{1}\mathbf{1}^\top$ denote the orthogonal projector onto $\mathbf{1}^\perp$, and let $\widetilde{L}_n = P_{\mathbf{1}^\perp} L_n P_{\mathbf{1}^\perp}\big|_{\mathbf{1}^\perp}$ be the restricted Laplacian on the (n^2-1)-dimensional zero-mean subspace. On $\mathbf{1}^\perp$, the irrational roots $\lambda_\pm(n)$ collapse into exact integers:*

$$\operatorname{spec}\big(\widetilde{L}_n\big|_{\mathbf{1}^\perp}\big) = \begin{cases} \left\{0^{[n^2-2n-1]},\ (n-2)^{[1]},\ n^{[2n-1]}\right\}, & \text{if } n \geq 4 \text{ is even}, \\ \left\{0^{[n^2-2n-1]},\ (n-1)^{[2]},\ n^{[2n-2]}\right\}, & \text{if } n \geq 3 \text{ is odd}. \end{cases} \tag{6}$$

Proof. Write $L_n = L_{\mathrm{rc}} + D$, where $L_{\mathrm{rc}} = \sum_{i=1}^{n} \mathbf{r}_i \mathbf{r}_i^\top + \sum_{j=1}^{n} \mathbf{c}_j \mathbf{c}_j^\top$ and $D = \mathbf{d}_+ \mathbf{d}_+^\top + \mathbf{d}_- \mathbf{d}_-^\top$. As the Cartesian product of two complete graphs $K_n \square K_n$, the operator L_{rc} has eigenvalues $2n$ (multiplicity 1, spanned by $\mathbf{1}$), n (multiplicity $2n-2$, spanned by zero-mean pure row functions $\mathcal{V}_{\mathrm{row}} = \{\mathbf{u} \otimes \mathbf{1}_n : \mathbf{u}^\top \mathbf{1}_n = 0\}$ and pure column functions $\mathcal{V}_{\mathrm{col}} = \{\mathbf{1}_n \otimes \mathbf{v} : \mathbf{v}^\top \mathbf{1}_n = 0\}$), and 0 (multiplicity $(n-1)^2$, the tensor-product subspace $\mathcal{V}_{\mathrm{row}} \otimes \mathcal{V}_{\mathrm{col}}$).

Because every row and every column intersects $\mathbf{d}_+$ and $\mathbf{d}_-$ in exactly one cell, $\mathbf{r}_i^\top \mathbf{d}_\pm = \mathbf{c}_j^\top \mathbf{d}_\pm = 1$ for all i, j. Thus for any zero-mean row or column mode $\mathbf{w} \in \mathcal{V}_{\mathrm{row}} \oplus \mathcal{V}_{\mathrm{col}} \subset \mathbf{1}^\perp$, we have $\mathbf{d}_+^\top \mathbf{w} = \mathbf{d}_-^\top \mathbf{w} = 0$. Consequently, the $(2n-2)$-dimensional subspace $\mathcal{V}_{\mathrm{row}} \oplus \mathcal{V}_{\mathrm{col}}$ is in the kernel of D and remains an exact eigenspace of both L_n and $\widetilde{L}_n$ with eigenvalue n and multiplicity $2n-2$.

It remains to analyze the action of L_n on the 3-dimensional subspace $\mathcal{S}_3 = \operatorname{span}\{\mathbf{1}, \mathbf{d}_+, \mathbf{d}_-\}$, which is orthogonal to $\mathcal{V}_{\mathrm{row}} \oplus \mathcal{V}_{\mathrm{col}}$. Introduce the symmetric and antisymmetric diagonal combinations $\mathbf{s}_d = \mathbf{d}_+ + \mathbf{d}_-$ and $\mathbf{a}_d = \mathbf{d}_+ - \mathbf{d}_-$. By symmetry, $\mathbf{1}^\top \mathbf{a}_d = 0$ and $L_{\mathrm{rc}} \mathbf{a}_d = 0$, so $\mathbf{a}_d \in \mathbf{1}^\perp$ is an exact eigenvector of both L_n and $\widetilde{L}_n$:

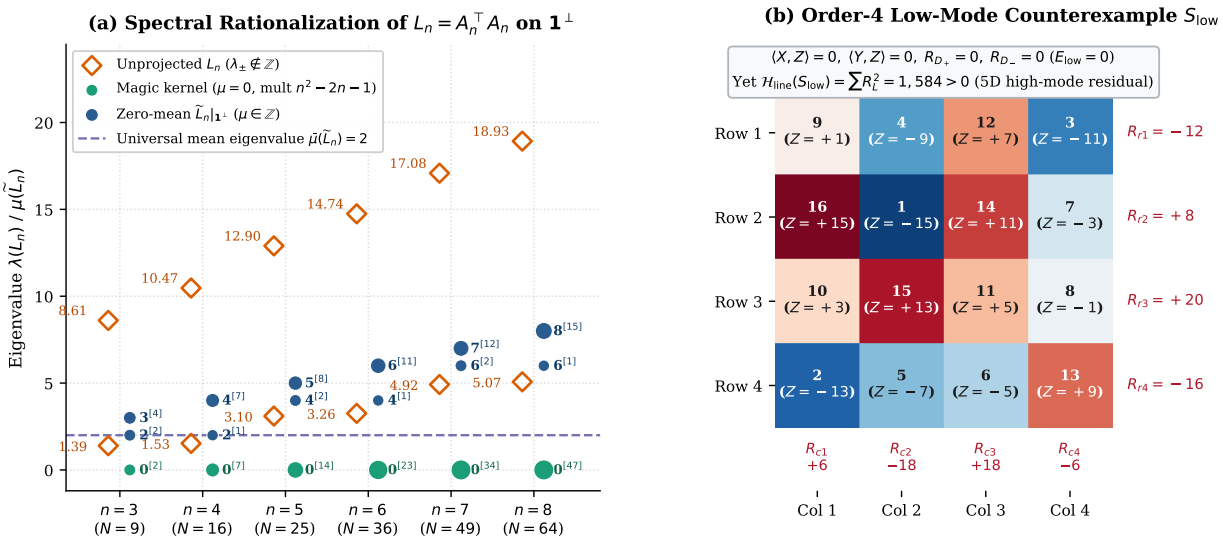


Figure 1: **Spectral rationalization of the line-incidence Laplacian $L_n = A_n^\top A_n$ and low-mode insufficiency.** **(a)** Comparison of the unprojected eigenvalues $\lambda(L_n)$ on $\mathbb{R}^{n^2}$ (hollow orange diamonds, showing the irrational branches $\lambda_\pm(n)$) against the zero-mean restricted spectrum $\mu(\widetilde{L}_n|_{\mathbf{1}^\perp})$ (filled navy and green circles, labeled $\mu^{[\text{mult}]}$) for orders $n = 3 \ldots 8$. Restricting to $\mathbf{1}^\perp$ collapses all eigenvalues into exact integers with universal mean $\bar{\mu}(\widetilde{L}_n) = 2$ (dashed purple line). **(b)** Explicit 4×4 low-mode counterexample S_{low} (Proposition 2.3) whose dipole moments $\langle X, Z \rangle = \langle Y, Z \rangle = 0$ and main diagonals $R_{D_\pm} = 0$ vanish ($E_{\text{low}} = 0$), yet whose high-frequency row and column residuals produce $\mathcal{H}_{\text{line}}(S_{\text{low}}) = 1{,}584 > 0$.

- For even n, $\mathbf{d}_+^\top \mathbf{d}_- = 0$, so $D\mathbf{a}_d = n\mathbf{a}_d$, contributing $+1$ to the multiplicity of eigenvalue n (giving $2n - 1$ in total). For the symmetric mode $\mathbf{s}_d$, since every row and column intersects $\mathbf{s}_d$ in 2 cells, $L_{\text{rc}}\mathbf{s}_d = 4\mathbf{1}$, while $L_{\text{rc}}\mathbf{1} = 2n\mathbf{1}$, $D\mathbf{1} = n\mathbf{s}_d$, and $D\mathbf{s}_d = n\mathbf{s}_d$. Thus the matrix of L_n on $\text{span}\{\mathbf{1}, \mathbf{s}_d\}$ is $\left[\begin{smallmatrix} 2n & 4 \\ n & n \end{smallmatrix}\right]$, whose characteristic polynomial is $\lambda^2 - 3n\lambda + 2n(n-2) = 0$, yielding $\lambda_\pm(n) = \frac{3n \pm \sqrt{n(n+16)}}{2}$. Projecting onto $\mathbf{1}^\perp$ replaces $\mathbf{s}_d$ by $\widetilde{\mathbf{s}}_d = \mathbf{s}_d - \frac{2}{n}\mathbf{1} \in \mathcal{V}_{\text{row}} \otimes \mathcal{V}_{\text{col}}$, on which $L_{\text{rc}}\widetilde{\mathbf{s}}_d = 0$ and $P_{\mathbf{1}^\perp} D\widetilde{\mathbf{s}}_d = (n-2)\widetilde{\mathbf{s}}_d$, yielding the exact integer eigenvalue $\mu = n - 2$ with multiplicity 1.

- For odd n, $\mathbf{d}_+^\top \mathbf{d}_- = 1$ (shared central cell), so $\|\mathbf{a}_d\|^2 = 2(n-1)$ and $D\mathbf{a}_d = (n-1)\mathbf{a}_d$, giving eigenvalue $n - 1$ for both L_n and $\widetilde{L}_n$. For the symmetric mode $\mathbf{s}_d$, $\mathbf{s}_d^\top \mathbf{s}_d = 2n + 2$, so $D\mathbf{s}_d = (n+1)\mathbf{s}_d$ and the 2×2 matrix of L_n on $\text{span}\{\mathbf{1}, \mathbf{s}_d\}$ is $\left[\begin{smallmatrix} 2n & 4 \\ n & n+1 \end{smallmatrix}\right]$ with characteristic polynomial $\lambda^2 - (3n+1)\lambda + 2n(n-1) = 0$, giving $\lambda_\pm(n) = \frac{3n+1 \pm \sqrt{n^2+14n+1}}{2}$. Projecting onto $\mathbf{1}^\perp$ leaves $\widetilde{\mathbf{s}}_d = \mathbf{s}_d - \frac{2}{n}\mathbf{1}$, on which $\widetilde{L}_n\widetilde{\mathbf{s}}_d = (n-1)\widetilde{\mathbf{s}}_d$, giving a second integer eigenvalue $\mu = n - 1$ (total multiplicity 2).

Finally, the orthogonal complement of $\text{span}\{\widetilde{\mathbf{s}}_d, \mathbf{a}_d\}$ inside the $(n-1)^2$-dimensional subspace $\mathcal{V}_{\text{row}} \otimes \mathcal{V}_{\text{col}}$ has dimension $(n-1)^2 - 2 = n^2 - 2n - 1$ and is annihilated by both L_{rc} and D, establishing Equation 4, Equation 5, and Equation 6. □

2.2 Universal Mean Eigenvalue and Random Permutation Ensemble Law

Theorem 2.1 reveals a remarkable parity-independent conservation law across all orders $n \geq 3$ (Table 2 and Figure 1a):

Theorem 2.2 (Universal Mean Eigenvalue and Ensemble Expectation)**.** *For every integer $n \geq 3$ (both odd and even), the trace of the zero-mean restricted Laplacian $\widetilde{L}_n$ on the (n^2-1)-dimensional hyperplane $\mathbf{1}^\perp$ is*

$$\text{tr}\left(\widetilde{L}_n\big|_{\mathbf{1}^\perp}\right) = 2(n^2 - 1), \tag{7}$$

Table 2: Exact spectral invariants of the unprojected Laplacian L_n on $\mathbb{R}^{n^2}$ and rationalized zero-mean Laplacian $\widetilde{L}_n$ on $\mathbf{1}^\perp$ across orders $n = 3 \dots 8$, verified by `verify_paper_d.py`.

n	$N = n^2$	$\operatorname{rank}(A_n)$	$\dim\ker(L_n)$	Irrational $\lambda_\pm(L_n)$	Zero-Mean $\operatorname{spec}(\widetilde{L}_n\|_{\mathbf{1}^\perp})$	$\operatorname{tr}(\widetilde{L}_n)$	$\mathbb{E}[\mathcal{H}_{\text{line}}(S)] = 2\|\mathbf{z}\|^2$
3	9	7	2	1.3944, 8.6056	$\{0^{[2]}, 2^{[2]}, 3^{[4]}\}$	16	480
4	16	9	7	1.5279, 10.4721	$\{0^{[7]}, 2^{[1]}, 4^{[7]}\}$	30	2,720
5	25	11	14	3.1010, 12.8990	$\{0^{[14]}, 4^{[2]}, 5^{[8]}\}$	48	10,400
6	36	13	23	3.2554, 14.7446	$\{0^{[23]}, 4^{[1]}, 6^{[11]}\}$	70	31,080
7	49	15	34	4.9172, 17.0828	$\{0^{[34]}, 6^{[2]}, 7^{[12]}\}$	96	78,400
8	64	17	47	5.0718, 18.9282	$\{0^{[47]}, 6^{[1]}, 8^{[15]}\}$	126	174,720

so the mean eigenvalue of $\widetilde{L}_n$ *on* $\mathbf{1}^\perp$ *is* ***identically*** 2*:*

$$\bar{\mu}(\widetilde{L}_n) = \frac{\operatorname{tr}(\widetilde{L}_n|_{\mathbf{1}^\perp})}{n^2 - 1} = 2. \tag{8}$$

Consequently, for a uniform random permutation $S \sim \mathrm{Unif}(S_{n^2})$ *with centred height vector* $\mathbf{z}(S)$ *of constant squared norm* $\|\mathbf{z}\|^2 = \frac{1}{3}n^2(n^4-1)$*, the expected line-indicator Hamiltonian is exactly twice the height variance sum:*

$$\mathbb{E}_{S\sim\mathrm{Unif}(S_{n^2})}\big[\mathcal{H}_{\text{line}}(S)\big] = \bar{\mu}(\widetilde{L}_n)\,\|\mathbf{z}\|^2 = 2\,\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4-1). \tag{9}$$

Proof. By Theorem 2.1, for even $n \geq 4$, summing the eigenvalues of $\widetilde{L}_n$ on $\mathbf{1}^\perp$ gives $(n-2)+(2n-1)n = 2n^2-2 = 2(n^2-1)$. For odd $n \geq 3$, the sum is $2(n-1)+(2n-2)n = 2(n-1)(n+1) = 2(n^2-1)$. Dividing by $\dim(\mathbf{1}^\perp) = n^2-1$ gives $\bar{\mu}(\widetilde{L}_n) = 2$ for every $n \geq 3$. Under the uniform distribution on S_{n^2}, the covariance matrix of $\mathbf{z}(S)$ is isotropic on $\mathbf{1}^\perp$: $\mathbb{E}[\mathbf{z}\mathbf{z}^\top] = \frac{\|\mathbf{z}\|^2}{n^2-1}P_{\mathbf{1}^\perp}$ [5]. Therefore $\mathbb{E}[\mathbf{z}^\top L_n \mathbf{z}] = \operatorname{tr}(L_n\mathbb{E}[\mathbf{z}\mathbf{z}^\top]) = \frac{\|\mathbf{z}\|^2}{n^2-1}\operatorname{tr}(\widetilde{L}_n|_{\mathbf{1}^\perp}) = 2\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4-1)$. □

2.3 Low-Mode Insufficiency and the $(2n-3)$-Dimensional High-Mode Subspace

In Paper II [10] and Paper III [9], we proved that semi-magic row and column balance implies vanishing dipole covariance $\langle X, Z\rangle = \langle Y, Z\rangle = 0$, so that every magic square has an isotropic inertia tensor. A natural physical question is whether the converse holds when supplemented by the two main diagonal constraints: if a permutation $S \in S_{n^2}$ satisfies zero dipole tilt ($\langle X, Z\rangle = \langle Y, Z\rangle = 0$) and zero diagonal residuals ($R_{D_+} = R_{D_-} = 0$), must S be a magic square? By Theorem 2.1, the answer is emphatically negative for all $n \geq 3$:

Proposition 2.3 (Low-Mode Insufficiency and Explicit Order-4 Counterexample)**.** *Define the* 4*-mode dipole-plus-diagonal energy functional*

$$E_{\text{low}}(S) = \langle X, Z\rangle^2 + \langle Y, Z\rangle^2 + R_{D_+}(S)^2 + R_{D_-}(S)^2. \tag{10}$$

For any order $n \geq 3$*, the four linear functionals* $(\langle X, \cdot\rangle, \langle Y, \cdot\rangle, R_{D_+}, R_{D_-})$ *span a* 4*-dimensional subspace of the* $(2n+1)$*-dimensional active image* $\operatorname{im}(\widetilde{L}_n) \subset \mathbf{1}^\perp$*. Its orthogonal complement within* $\operatorname{im}(\widetilde{L}_n)$ *has dimension*

$$\dim\big(\operatorname{im}(\widetilde{L}_n) \cap \ker(E_{\text{low}})\big) = (2n+1) - 4 = 2n - 3, \tag{11}$$

spanned by nonlinear row and column polynomial modes $\langle X^k, Z\rangle$ *and* $\langle Y^k, Z\rangle$ *for* $2 \leq k \leq n-1$*. At order* $n = 4$ *(*$2n-3 = 5$*), the explicit permutation*

$$S_{\text{low}} = \begin{bmatrix} 9 & 4 & 12 & 3 \\ 16 & 1 & 14 & 7 \\ 10 & 15 & 11 & 8 \\ 2 & 5 & 6 & 13 \end{bmatrix} \in S_{16} \tag{12}$$

Table 3: Multiscale window entropy $\widetilde{H}_p(2)$, interior and toroidal 2×2 quartet energies $W_{2\times 2}$, complement breaks B_{comp}, and mixed octupole moment M_{xyz} across constructive magic square families ($n = 3\ldots 8$).

Family	Order / Class	$\widetilde{H}_1(2)$	$\widetilde{H}_2(2)$	$W^{\text{int}}_{2\times 2}(S)$	$W^{\text{tor}}_{2\times 2}(S)$	Breaks B_{comp}	Octupole M_{xyz}
Lo Shu (Siamese)	$n = 3$ (odd)	0.99435	0.99073	80	240	0/4	0
Doubly-Even Mask	$n = 4$ ($4k$)	0.97245	0.91498	544	1,088	0/8	0
Siamese (de La Loubère)	$n = 5$ (odd)	0.97438	0.94254	16,320	25,200	0/12	0
Conway LUX Method	$n = 6$ ($4k+2$)	0.97011	0.93161	84,640	97,440	18/18	-132
Doubly-Even Mask	$n = 8$ ($4k$)	0.96123	0.90598	112,320	149,760	0/32	0

satisfies $\langle X, Z\rangle = 0$, $\langle Y, Z\rangle = 0$, $R_{D_+}(S_{\text{low}}) = 0$, *and* $R_{D_-}(S_{\text{low}}) = 0$ ($E_{\text{low}}(S_{\text{low}}) = 0$), *yet exhibits non-zero row residuals* $\mathbf{R}_{\text{row}} = (-12, +8, +20, -16)$, *column residuals* $\mathbf{R}_{\text{col}} = (+6, -18, +18, -6)$, *and total line energy* $\mathcal{H}_{\text{line}}(S_{\text{low}}) = 1{,}584 > 0$ (*Figure 1b*).

Proof. Because $\langle X, Z\rangle = \sum_{j=1}^{n} X_j R_{c_j}(S)$ and $\langle Y, Z\rangle = \sum_{i=1}^{n} Y_i R_{r_i}(S)$ are linear combinations of the column and row residuals, $\text{span}\{X, Y, \widetilde{\mathbf{d}}_+, \widetilde{\mathbf{d}}_-\}$ is a 4-dimensional subspace of $\text{im}(\widetilde{L}_n)$. Since $\dim \text{im}(\widetilde{L}_n) = 2n + 1$ by Theorem 2.1, the high-frequency residual subspace inside $\text{im}(\widetilde{L}_n)$ has dimension $(2n + 1) - 4 = 2n - 3$. Direct evaluation on S_{low} in Equation 12 yields row sums $(28, 38, 44, 26)$ (centred residuals $(-12, 8, 20, -16)$), column sums $(37, 25, 43, 31)$ (centred residuals $(6, -18, 18, -6)$), and diagonal sums $(34, 34)$ (centred residuals $(0, 0)$). Taking inner products with $X = (-3, -1, 1, 3)$ and $Y = (3, 1, -1, -3)$ gives $\langle X, Z\rangle = -3(6) - 1(-18) + 1(18) + 3(-6) = 0$ and $\langle Y, Z\rangle = 3(-12) + 1(8) - 1(20) - 3(-16) = 0$, while $\mathcal{H}_{\text{line}}(S_{\text{low}}) = (-12)^2 + 8^2 + 20^2 + (-16)^2 + 6^2 + (-18)^2 + 18^2 + (-6)^2 = 1{,}584$. □

3 Multiscale Window Entropies and 2×2 Quartet Equivalence

Once a permutation reaches the magic ground-state manifold $\ker(\widetilde{L}_n)$ where $\mathcal{H}_{\text{line}}(S) = 0$, all $2n+2$ global 1D line sums are identical. However, distinct magic squares of the same order exhibit vastly different local 2D spatial organization [6, 14, 3]. To resolve this internal structure across scales, we introduce sliding $k \times k$ sub-window energies and Shannon entropies, and evaluate them exhaustively across all 880 Frénicle order-4 magic squares and higher-order constructive families.

3.1 Sliding-Window Shannon Entropy and Constructive Family Scaling

For any scale $k \in \{1, \ldots, n\}$, an $n \times n$ grid contains $M_k = (n - k + 1)^2$ interior contiguous $k \times k$ sub-windows $W^{(k)}_{r,c} = \{r, \ldots, r + k - 1\} \times \{c, \ldots, c + k - 1\}$ $(1 \le r, c \le n - k + 1)$. For any exponent $p \in \{1, 2, 4\}$, define the local ℓ^p window weight and normalized Shannon window entropy by

$$w_p\big(W^{(k)}_{r,c}\big) = \sum_{(i,j)\in W^{(k)}_{r,c}} |Z_{ij}|^p, \qquad p^{(k)}_{r,c} = \frac{w_p(W^{(k)}_{r,c})}{\sum_{r',c'} w_p(W^{(k)}_{r',c'})}, \qquad \widetilde{H}_p(k) = \frac{-\sum_{r,c} p^{(k)}_{r,c} \log_2 p^{(k)}_{r,c}}{\log_2\big((n-k+1)^2\big)}. \quad (13)$$

Here $\widetilde{H}_p(k) \in (0, 1]$ measures the spatial uniformity of height magnitude across all $k \times k$ sub-windows ($\widetilde{H}_p(k) = 1$ if and only if every $k \times k$ window carries identical ℓ^p weight).

Figure 2a and Table 3 compare the multiscale entropy profile $\widetilde{H}_p(k)$ across five classical constructive families: odd-order de La Loubère (Siamese) squares ($n = 3, 5$) [4], doubly-even 4×4 diagonal-mask squares ($n = 4, 8$) [1], and Conway's LUX singly-even construction ($n = 6$) [2]. At $k = 2$, the 3×3 Lo Shu square achieves the highest local uniformity ($\widetilde{H}_1(2) = 0.99435$, $\widetilde{H}_2(2) = 0.99073$), whereas doubly-even mask constructions at $n = 4, 8$ concentrate extreme values symmetrically in corner and central blocks, lowering $\widetilde{H}_2(2)$ to 0.91498 ($n = 4$) and 0.90598 ($n = 8$). Crucially, while $n = 3, 4, 5, 8$ constructions are centrosymmetric ($B_{\text{comp}} = 0$ complement breaks and $M_{xyz} = 0$), Conway's $n = 6$ LUX method breaks all 18/18 antipodal complement pairs and generates a non-zero mixed octupole moment $M_{xyz} = -132$.

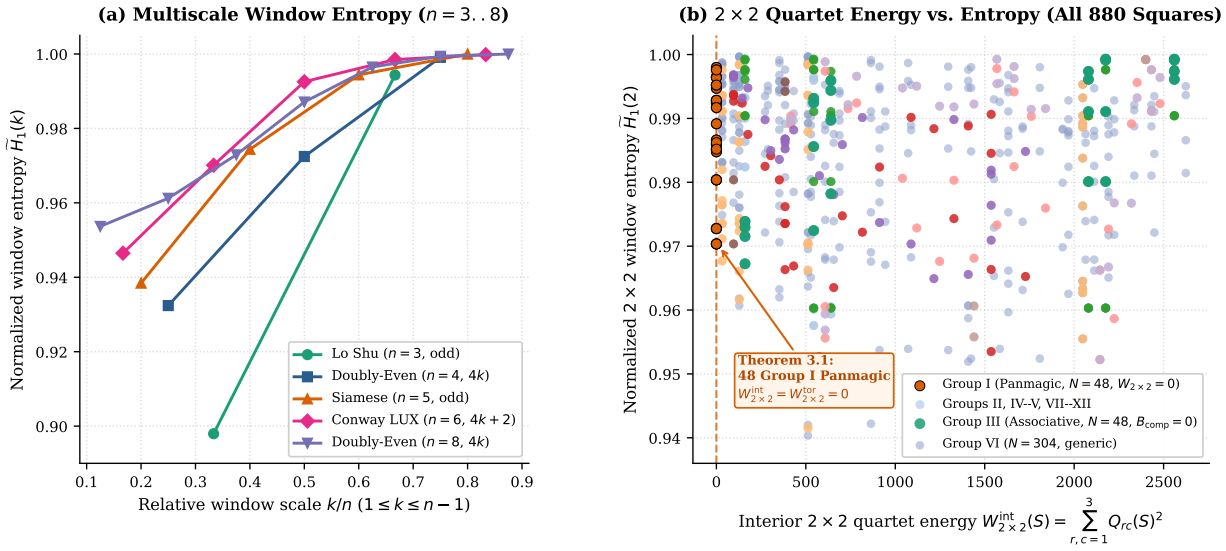


Figure 2: **Multiscale sliding-window Shannon entropy and 2×2 quartet energy census. (a)** Normalized ℓ_1 window entropy $\widetilde{H}_1(k)$ versus relative scale k/n across five constructive magic-square families ($n = 3$ Lo Shu, $n = 4, 8$ Doubly-Even, $n = 5$ Siamese, and $n = 6$ Conway LUX). **(b)** Joint distribution of interior 2×2 quartet residual energy $W^{\text{int}}_{2\times 2}(S)$ and window entropy $\widetilde{H}_1(2)$ across all 880 Frénicle order-4 magic squares partitioned by Dudeney Groups I–XII, illustrating Theorem 3.1: $W^{\text{int}}_{2\times 2}(S) = 0$ holds if and only if $W^{\text{tor}}_{2\times 2}(S) = 0$, isolating the 48 panmagic squares of Group I.

3.2 Equivalence of Interior and Toroidal 2×2 Quartet Balance at Order 4

Beyond magnitude entropy, the signed sum of each 2×2 quartet $Q_{r,c}(S) = Z_{r,c} + Z_{r+1,c} + Z_{r,c+1} + Z_{r+1,c+1}$ (with indices taken modulo n in the toroidal case) defines the *interior* and *toroidal* 2×2 *quartet energies*:

$$W^{\text{int}}_{2\times 2}(S) = \sum_{r=1}^{n-1}\sum_{c=1}^{n-1} Q_{r,c}(S)^2, \qquad W^{\text{tor}}_{2\times 2}(S) = \sum_{r=1}^{n}\sum_{c=1}^{n} Q_{r,c}(S)^2. \tag{14}$$

At order $n = 4$, $W^{\text{int}}_{2\times 2}$ sums over the 9 interior 2×2 blocks, whereas $W^{\text{tor}}_{2\times 2}$ includes the 7 boundary-wrapping blocks. We prove that on the order-4 magic manifold, interior quartet balance automatically implies full toroidal quartet balance:

Theorem 3.1 (Interior–Toroidal 2×2 Quartet Equivalence)**.** *Let S be any magic square of order $n = 4$. Then its 9 interior 2×2 blocks have zero centred sum ($W^{\text{int}}_{2\times 2}(S) = 0$) **if and only if** all 16 toroidal 2×2 blocks have zero centred sum ($W^{\text{tor}}_{2\times 2}(S) = 0$). Across the 880 Frénicle magic squares of order 4, this condition holds on **exactly the 48 panmagic (diabolic) squares of Dudeney Group I** (Figure 2b and Table 4).*

Proof. Clearly $W^{\text{tor}}_{2\times 2}(S) = 0 \implies W^{\text{int}}_{2\times 2}(S) = 0$. Conversely, suppose S is an order-4 magic square ($\mathcal{H}_{\text{line}}(S) = 0$) with $Q_{r,c}(S) = 0$ for all $1 \le r, c \le 3$. For any horizontally wrapping 2×2 block $Q_{r,4}(S)$ $(1 \le r \le 3)$, the sum of rows r and $r + 1$ vanishes by magic row balance:

$$0 = R_r(S) + R_{r+1}(S) = Q_{r,1}(S) + Q_{r,3}(S) = Q_{r,2}(S) + Q_{r,4}(S). \tag{15}$$

Since $Q_{r,2}(S) = 0$ by interior balance, Equation 15 forces $Q_{r,4}(S) = -Q_{r,2}(S) = 0$ for $r \in \{1, 2, 3\}$. Symmetrically, summing columns c and $c + 1$ gives $0 = R_{c_c}(S) + R_{c_{c+1}}(S) = Q_{2,c}(S) + Q_{4,c}(S)$, so $Q_{2,c}(S) = 0 \implies Q_{4,c}(S) = 0$ for $c \in \{1, 2, 3\}$. Finally, the corner-wrapping quartet $Q_{4,4}(S) = Z_{4,4} + Z_{1,4} + Z_{4,1} + Z_{1,1}$ satisfies $Q_{4,4}(S) = (R_{r_1}(S) + R_{r_4}(S)) - Q_{4,2}(S) = 0 - 0 = 0$. Thus all 16 toroidal quartets vanish. It is a classical theorem of Frénicle and Dudeney [6, 14] that a 4×4 magic square has all 2×2 toroidal quartets equal to $2M_4 = 34$ ($Q_{r,c} = 0$) if and only if it is panmagic, which characterizes the 48 squares of Dudeney Group I. □

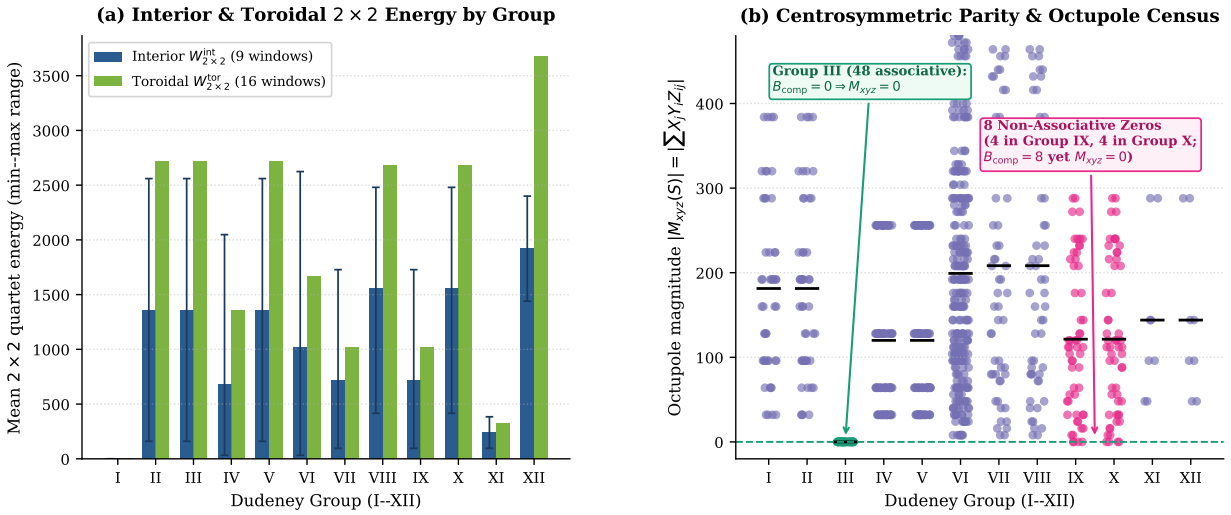


Figure 3: **Group-resolved 2×2 quartet energies and mixed octupole census across Dudeney Groups I–XII. (a)** Mean interior $W^{\text{int}}_{2\times2}$ (9 windows, navy bars with min–max range) and toroidal $W^{\text{tor}}_{2\times2}$ (16 windows, green bars) across the 12 Dudeney groups. **(b)** Magnitude of the mixed octupole moment $|M_{xyz}(S)| = |\sum X_j Y_i Z_{ij}|$ across all 880 squares (Proposition 3.2), highlighting the 48 associative squares of Group III ($B_{\text{comp}} = 0 \implies M_{xyz} = 0$) and the 8 anomalous non-associative octupole-zero squares in Groups IX and X ($B_{\text{comp}} = 8$ yet $M_{xyz} = 0$).

3.3 Centrosymmetric Complement Parity and the Anomalous Octupole Zeros

In Paper II [10] and Paper III [9], the mixed third-order moment (octupole) $M_{xyz}(S) = \sum_{i,j=1}^{n} X_j Y_i Z_{ij}$ governed the asymmetry between static and dynamic axes. To connect $M_{xyz}(S)$ to discrete cell pairing, define the *complement-break count* $B_{\text{comp}}(S)$ as the number of antipodal cell pairs $\{(i,j),(n+1-i,n+1-j)\}$ that fail the associative condition $Z_{ij} + Z_{n+1-i,n+1-j} = 0$ ($s_{ij} + s_{n+1-i,n+1-j} = n^2 + 1$).

Proposition 3.2 (Binary Complement Breaking and Octupole Zero Census)**.** *Across all* 880 *Frénicle magic squares of order* 4*:*

1. ***Binary Complement Breaking:*** *The complement-break count* $B_{\text{comp}}(S) \in \{0, 1, \ldots, 8\}$ *takes* ***only the two extreme values*** $\{0, 8\}$*:* $B_{\text{comp}}(S) = 0$ *on the* 48 *associative (symmetric) squares of Dudeney Group III, and* $B_{\text{comp}}(S) = 8$ *on all remaining* 832 *squares (no order-*4 *magic square has* $1 \le B_{\text{comp}}(S) \le 7$*).*

2. ***Symmetry-Protected vs. Accidental Octupole Zeros:*** *The mixed octupole moment* $M_{xyz}(S) = \sum_{i,j} X_j Y_i Z_{ij}$ *vanishes on* ***exactly*** 56 ***of the*** 880 ***squares*** *(Figure 3b and Table 4):*

   - *All* 48 *associative squares of Dudeney Group III* ($B_{\text{comp}}(S) = 0$*), where inversion parity* $Z_{5-i,5-j} = -Z_{ij}$ *combined with* $X_{5-j}Y_{5-i} = (-X_j)(-Y_i) = +X_j Y_i$ *forces each antipodal pair to cancel:* $X_j Y_i (Z_{ij} + Z_{5-i,5-j}) = 0$*.*
   - *Exactly* 8 *non-associative squares* ($B_{\text{comp}}(S) = 8$*)—*4 *in Dudeney Group IX and* 4 *in Dudeney Group X—where non-zero antipodal pair sums* $Z_{ij} + Z_{5-i,5-j} \neq 0$ *cancel algebraically across the* 8 *pairs.*

Across all 880 squares, the global maximum of 2×2 window entropy is achieved by Frénicle square `#123` in Group VI ($\widetilde{H}_1(2) = 0.999669$), while the global minimum occurs at square `#599` in Group VI ($\widetilde{H}_1(2) = 0.940345$). Notice also in Table 4 and Figure 3a that Groups II, III, and V share the exact same mean interior quartet energy $\langle W^{\text{int}}_{2\times2} \rangle = 1{,}360 = \|\mathbf{z}\|^2$ and mean toroidal quartet energy $\langle W^{\text{tor}}_{2\times2} \rangle = 2{,}720 = 2\|\mathbf{z}\|^2$.

Table 4: Complete multiscale window and multipole census across the 12 Dudeney groups of the 880 order-4 magic squares, verified by `verify_paper_d.py`.

Dudeney Group	N_g	B_{comp}	Mean $\widetilde{H}_1(2)$	Mean $\widetilde{H}_2(2)$	$W^{\text{int}}_{2\times2}$ (mean [min, max])	$W^{\text{tor}}_{2\times2}$ (mean)	$\lvert M_{xyz}\rvert$ (mean, $\# = 0$)
I (Panmagic)	48	8	0.98722	0.96742	0.0 [0, 0]	0.0	181.3 (0/48)
II (Semi-pan.)	48	8	0.98148	0.95260	1360.0 [160, 2560]	2720.0	181.3 (0/48)
III (Assoc.)	48	0	0.98807	0.96789	1360.0 [160, 2560]	2720.0	0.0 (48/48)
IV	96	8	0.97919	0.94745	680.0 [32, 2048]	1360.0	120.0 (0/96)
V	96	8	0.98831	0.96951	1360.0 [160, 2560]	2720.0	120.0 (0/96)
VI	304	8	0.98356	0.95870	1017.7 [32, 2624]	1671.2	199.2 (0/304)
VII	56	8	0.97874	0.94983	720.6 [96, 1728]	1015.4	208.3 (0/56)
VIII	56	8	0.98127	0.95081	1554.9 [416, 2480]	2684.0	208.3 (0/56)
IX	56	8	0.98432	0.95886	720.6 [96, 1728]	1015.4	121.4 (4/56)
X	56	8	0.98705	0.96844	1554.9 [416, 2480]	2684.0	121.4 (4/56)
XI	8	8	0.98516	0.96109	240.0 [96, 384]	320.0	144.0 (0/8)
XII	8	8	0.97871	0.94441	1920.0 [1440, 2400]	3680.0	144.0 (0/8)

4 Combinatorial Swap Landscape and Antipodal Traps

Whereas Section 2 showed that the continuous quadratic form $\mathbf{z}^\top L_n \mathbf{z}$ on $\mathbf{1}^\perp$ is convex and barrier-free, restricting $\mathbf{z}$ to the discrete permutation polytope S_{n^2} induces a rugged combinatorial energy landscape under elementary cell swaps. We now derive the exact algebraic law governing 2-cycle transpositions and perform an exhaustive basin census of S_9 ($n = 3$) and a large-scale relaxation census of S_{16} ($n = 4$).

4.1 Exact Quadratic Transposition Law

Let $\mathbf{g}(S) = \frac{1}{2}\nabla\mathcal{H}_{\text{line}}(\mathbf{z}) = L_n\mathbf{z} = A_n^\top \mathbf{R}(S) \in \mathbb{Z}^{n^2}$ denote the *cell-stress vector*, whose component $g_a(S) = \sum_{L\ni a} R_L(S)$ at cell $a \in \{1, \dots, n^2\}$ is the sum of the centred residuals of the 2, 3, or 4 magic lines passing through cell a. Swapping the values at cells a and b via the transposition $\tau_{ab} \in S_{n^2}$ replaces $\mathbf{z}$ by $\mathbf{z}' = \mathbf{z} - d(\mathbf{e}_a - \mathbf{e}_b)$, where $d = Z_a - Z_b = 2(s_a - s_b)$. Expanding $\mathbf{z}'^\top L_n \mathbf{z}'$ yields:

Proposition 4.1 (Exact Transposition Energy Formula)**.** *For any permutation $S \in S_{n^2}$ and any pair of cells $a \neq b$ with height difference $d = Z_a - Z_b$, the change in the line-indicator Hamiltonian under the transposition τ_{ab} is*

$$\Delta\mathcal{H}_{\text{line}}(\tau_{ab}) = \mathcal{H}_{\text{line}}(\tau_{ab}S) - \mathcal{H}_{\text{line}}(S) = -2d\big(g_a(S) - g_b(S)\big) + c_{ab}\, d^2, \tag{16}$$

where $c_{ab} = (L_n)_{aa} + (L_n)_{bb} - 2(L_n)_{ab} = \|A_n(\mathbf{e}_a - \mathbf{e}_b)\|_2^2 \in \{2, 3, 4, 5, 6, 7, 8\}$ counts the number of magic lines in $\mathcal{L}_n$ that pass through exactly one *of the two cells a and b. In particular, $\Delta\mathcal{H}_{\text{line}}(\tau_{ab}) < 0$ if and only if $(g_a - g_b)/d > c_{ab}/2$.*

Proposition 4.1 explains why discrete 2-cycle descent becomes frustrated near the ground state: when line residuals $R_L(S)$ are small (±2 in centred units, or ±1 in standard units), the linear stress release $-2d(g_a - g_b)$ has magnitude $O(|d|)$, whereas the quadratic self-energy penalty $c_{ab}d^2 \geq 8$ (since $|d| \geq 2$ and $c_{ab} \geq 2$) grows quadratically with the height jump $|d|$ and blocks any single transposition from lowering $\mathcal{H}_{\text{line}}$.

4.2 Exhaustive S_9 Landscape Census and the 8 Antipodal Traps of Order 3

At order $n = 3$, the symmetric group S_9 has $9! = 362{,}880$ states, permitting an exact, exhaustive classification of the entire combinatorial energy landscape (Theorem 4.2 and Figure 4):

Theorem 4.2 (Exhaustive S_9 Landscape and the 8 Strict Local Minima of Order 3)**.** *Across all $9! = 362{,}880$ permutations of S_9 under the line-indicator Hamiltonian $\mathcal{H}_{\text{line}}(S)$:*

1. ***Complete Classification of Local Minima:*** *There are* ***exactly*** 16 ***local minima*** *in S_9 under the $\binom{9}{2} = 36$ elementary 2-cycle transpositions, all 16 of which are strict local minima ($\Delta\mathcal{H}_{\text{line}}(\tau_{ab}) > 0$ for all $a \neq b$):*

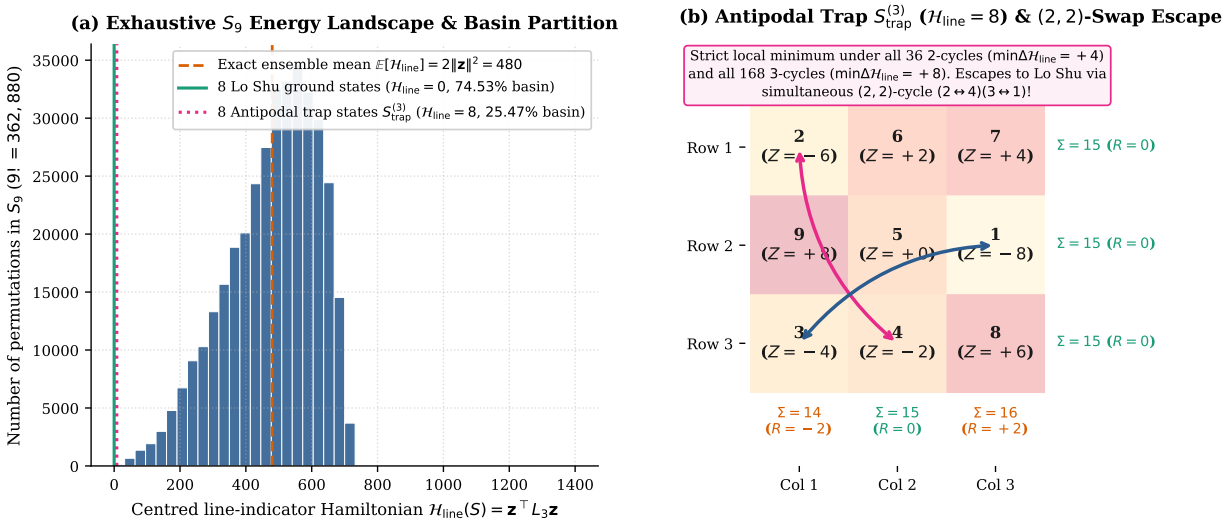


Figure 4: **Exhaustive S_9 permutation landscape (362,880 states) and the 8 antipodal traps at order $n = 3$. (a)** Exact histogram of the centred line Hamiltonian $\mathcal{H}_{\text{line}}(S)$ across all $9! = 362{,}880$ states of S_9, showing the ensemble mean $\mathbb{E}[\mathcal{H}_{\text{line}}] = 480$ (Theorem 2.2), the 8 Lo Shu ground states at $\mathcal{H}_{\text{line}} = 0$, and the 8 strict local-minimum traps $S^{(3)}_{\text{trap}}$ at $\mathcal{H}_{\text{line}} = 8$. **(b)** Structure of the canonical trap $S^{(3)}_{\text{trap}}$ (Theorem 4.2): all 3 rows, the middle column, and both main diagonals sum to 15 ($R = 0$), while the outer columns sum to 14 ($R = -2$) and 16 ($R = +2$). Every 2-cycle ($\min \Delta\mathcal{H}_{\text{line}} = +4$) and every 3-cycle ($\min \Delta\mathcal{H}_{\text{line}} = +8$) strictly increases energy; escaping to the Lo Shu ground state requires the simultaneous $(2, 2)$ double swap $(2 \leftrightarrow 4)(3 \leftrightarrow 1)$.

- *The 8* ***global ground states*** *($\mathcal{H}_{\text{line}} = 0$), forming the single dihedral orbit $D_4 \cdot S_{\text{LoShu}}$.*
- ***Exactly** 8 **spurious strict local minima** at $\mathcal{H}_{\text{line}} = 8$ ($E_{\text{std}} = 2$), forming the single dihedral D_4 orbit of the antipodal trap*

$$S^{(3)}_{\text{trap}} = \begin{bmatrix} 2 & 6 & 7 \\ 9 & 5 & 1 \\ 3 & 4 & 8 \end{bmatrix}, \qquad \mathbf{Z}\big(S^{(3)}_{\text{trap}}\big) = \begin{bmatrix} -6 & +2 & +4 \\ +8 & 0 & -8 \\ -4 & -2 & +6 \end{bmatrix}, \qquad \mathcal{H}_{\text{line}}\big(S^{(3)}_{\text{trap}}\big) = 8. \tag{17}$$

2. ***Instability Under** 2**-Cycles and** 3**-Cycles, and** $(2,2)$**-Cycle Escape:** In $S^{(3)}_{\text{trap}}$, six of the eight magic lines (all three rows, the middle column, and both main diagonals) have zero residual ($R_L = 0$), while the two outer columns have residuals $R_{c_1} = -2$ ($\Sigma = 14$) and $R_{c_3} = +2$ ($\Sigma = 16$). Every single 2-cycle transposition strictly increases $\mathcal{H}_{\text{line}}$ by at least $+4$ ($\min_{\tau_{ab}} \Delta\mathcal{H}_{\text{line}} = +4$), **and every single 3-cycle permutation among all** $2\binom{9}{3} = 168$ **directed 3-cycles also strictly increases $\mathcal{H}_{\text{line}}$ by at least** $+8$. Escaping $S^{(3)}_{\text{trap}}$ to the Lo Shu ground state requires a simultaneous disjoint double transposition ($(2,2)$-cycle), such as $(2 \leftrightarrow 4)(3 \leftrightarrow 1)$ or $(6 \leftrightarrow 4)(7 \leftrightarrow 9)$.*

3. ***Exact Steepest-Descent Basin Partition:** Under deterministic 2-cycle steepest descent (choosing the lexicographically first pair (a, b) minimizing $\Delta\mathcal{H}_{\text{line}}(\tau_{ab})$), 270,442 **of the** 362,880 **states** (74.5266%) **converge to the** 8 **Lo Shu ground states**, while the remaining 92,438 **states** (25.4734%) **are captured by the** 8 **antipodal traps** $D_4 \cdot S^{(3)}_{\text{trap}}$.*

Proof. All 362,880 permutations of S_9, their 36 transposition deltas $\Delta\mathcal{H}_{\text{line}}(\tau_{ab})$ via Equation 16, all 168 directed 3-cycle deltas on trapped states, and the steepest-descent attractor map are evaluated exhaustively in `verify_paper_d.py` (Figure 4). To see algebraically why $S^{(3)}_{\text{trap}}$ in Equation 17 is immune to both 2-cycles and 3-cycles: in standard units $s_{ij} - 5 \in \{-4, \ldots, 4\}$, $E_{\text{std}}(S^{(3)}_{\text{trap}}) = (-1)^2 + (+1)^2 = 2$. Reducing E_{std} below 2 requires reaching a state where at most one line has residual ± 1 (impossible by parity) or reaching $E_{\text{std}} = 0$ directly. Because every row sum of $S^{(3)}_{\text{trap}}$

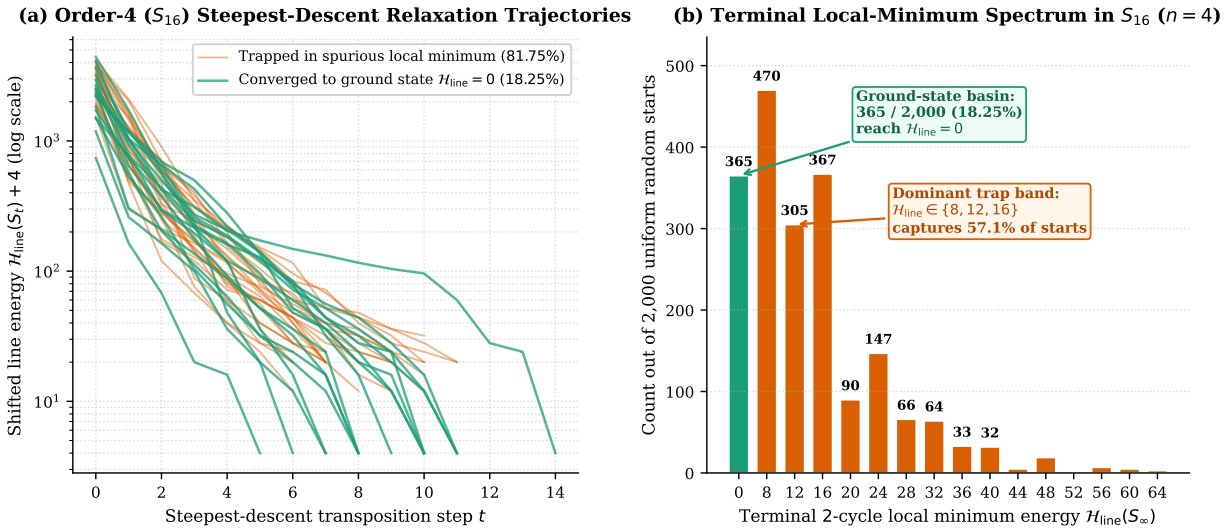


Figure 5: **Order-4 (S_{16}) steepest-descent relaxation trajectories and local-minimum spectrum (2,000 trials). (a)** Representative steepest-descent trajectories of $\mathcal{H}_{\text{line}}(S_t) + 4$ (log scale) versus transposition step t, contrasting trajectories that reach the magic ground state $\mathcal{H}_{\text{line}} = 0$ (green, 18.25%) against trajectories trapped in frustrated local minima (orange, 81.75%). **(b)** Histogram of terminal 2-cycle local-minimum energies $\mathcal{H}_{\text{line}}(S_\infty)$ across 2,000 uniform random starts in S_{16}. While 365/2,000 (18.25%) reach $\mathcal{H}_{\text{line}} = 0$, 57.1% of all starts become trapped in the low-energy frustration band $\mathcal{H}_{\text{line}} \in \{8, 12, 16\}$.

is already 15, any single 2-cycle between different rows alters two row sums by $\pm d \neq 0$ (adding $2d^2 \geq 2$ to the row energy), while any within-row 2-cycle changes the diagonal sums by at least ± 2. Similarly, any 3-cycle either touches two cells in one row and one cell in another row (breaking both row sums) or touches one cell in each of the three rows (preserving row sums only if all three swapped values are identical, which is impossible). By contrast, a simultaneous disjoint $(2, 2)$ double transposition such as $(2 \leftrightarrow 4)(3 \leftrightarrow 1)$ exchanges (-1) and $(+1)$ across two rows simultaneously, lowering E_{std} and unlocking direct descent to the Lo Shu ground state. □

4.3 Order-4 (S_{16}) Steepest-Descent Basin Census

At order $n = 4$, the state space S_{16} grows to $16! \approx 2.09 \times 10^{13}$ permutations, while the number of distinct magic ground states is $880 \times 8 = 7{,}040$. To quantify local-minimum trapping in S_{16}, we performed 2,000 independent steepest-descent trials starting from uniform random permutations (Figure 5 and Table 5).

As shown in Figure 5b and Table 5, pure 2-cycle steepest descent in S_{16} reaches a magic ground state ($\mathcal{H}_{\text{line}} = 0$) in only 365/2,000 trials (18.25%), terminating in a mean of 8.18 steps at a mean residual energy $\langle \mathcal{H}_{\text{line}}(S_\infty) \rangle = 14.43$ (median 12.0). The trapped configurations cluster heavily in the lowest frustration band $\mathcal{H}_{\text{line}} \in \{8, 12, 16\}$, which accounts for $470 + 305 + 367 = 1{,}142/2{,}000$ (57.1%) of all random starts. Testing all $2\binom{16}{3} = 1{,}120$ directed 3-cycles on each of the 1,635 trapped 2-cycle local minima reveals that 56 (3.43%) immediately escape via a single 3-cycle, motivating the hybrid continuous-discrete solver developed in Section 5.

5 Continuous "Bubble Trinity" Langevin Relaxation and Hybrid Settling

To overcome the discrete local-minimum traps identified in Section 4, the *Bubble Trinity* architecture in `physicalSolver` [12] relaxes the rigid permutation assignment by embedding the n^2 numbers as mobile viscoelastic particles ("bubbles") in continuous $\mathbb{R}^2$ plane coordinates $\mathbf{r}_a(t) = (x_a(t), y_a(t)) \in \mathbb{R}^2$ $(a = 1, \ldots, n^2)$ anchored near the lattice cell centers $\mathbf{r}_a^{(0)} = (c_a - \frac{n+1}{2}, r_a - \frac{n+1}{2})$.

Table 5: Summary of combinatorial swap-landscape invariants for S_9 ($n = 3$, exhaustive 362,880 states) and S_{16} ($n = 4$, 2,000 random starts), verified by `verify_paper_d.py`.

Landscape Observable / Metric	Order $n = 3$ (S_9, Exhaustive)	Order $n = 4$ (S_{16}, 2,000 Trials)
Total permutation state space $n^2!$	362,880	20,922,789,888,000
Magic ground states ($\mathcal{H}_{\text{line}} = 0$)	8 ($1 \times D_4$ orbit)	7,040 ($880 \times D_4$ orbits)
Ensemble mean energy $\mathbb{E}[\mathcal{H}_{\text{line}}(S)] = 2\|\mathbf{z}\|^2$	480	2,720
Elementary 2-cycle neighborhood size $\binom{n^2}{2}$	36	120
Directed 3-cycle neighborhood size $2\binom{n^2}{3}$	168	1,120
Spurious 2-cycle local minima ($\mathcal{H}_{\text{line}} > 0$)	Exactly 8 (at $\mathcal{H}_{\text{line}} = 8$)	Abundant ($\mathcal{H}_{\text{line}} \in \{8, 12, 16, \dots\}$)
2-cycle steepest-descent ground-state basin	270,442/362,880 (74.53%)	365/2,000 (18.25%)
Mean terminal $\mathcal{H}_{\text{line}}(S_\infty)$ under 2-cycle descent	2.04	14.43 (median 12.0)

5.1 Continuous Kernel-Induced Line Field and Langevin Equations of Motion

At any continuous configuration $\{\mathbf{r}_a(t)\}_{a=1}^{n^2}$, each lattice cell $b \in \{1, \dots, n^2\}$ samples a soft Gaussian kernel interpolation of the bubble heights Z_a:

$$W_{ab}(t) = \frac{\exp\big(-2\|\mathbf{r}_a(t) - \mathbf{r}_b^{(0)}\|_2^2\big)}{\sum_{a'=1}^{n^2} \exp\big(-2\|\mathbf{r}_{a'}(t) - \mathbf{r}_b^{(0)}\|_2^2\big)}, \qquad \widetilde{\mathbf{z}}(t) = W(t)^\top \mathbf{z}, \qquad \mathcal{H}_{\text{ind}}(t) = \|A_n \widetilde{\mathbf{z}}(t)\|_2^2. \tag{18}$$

When every bubble sits at its home lattice site ($\mathbf{r}_a = \mathbf{r}_a^{(0)}$), $\mathcal{H}_{\text{ind}}(t)$ approaches the discrete line Hamiltonian $\mathcal{H}_{\text{line}}(S)$. Away from the lattice, the continuous cell-stress vector $\widetilde{\mathbf{g}}(t) = A_n^\top A_n \widetilde{\mathbf{z}}(t)$ defines a smooth target attraction field $\mathbf{r}_a^{\text{target}}(t) = \sum_{b=1}^{n^2} \sigma_{ab}(t) \mathbf{r}_b^{(0)}$ via the Boltzmann-weighted stress pairing $\sigma_{ab}(t) \propto \exp(-Z_a \widetilde{g}_b(t)/T_{\text{field}})$, which pulls positive-height bubbles ($Z_a > 0$) toward cells with negative line residual ($\widetilde{g}_b < 0$) and vice versa. Each bubble evolves under the damped Langevin equation

$$\mathbf{v}_a(t + \Delta t) = \gamma_m \mathbf{v}_a(t) + \Delta t \Big[\mathbf{F}_a^{\text{well}} + \mathbf{F}_a^{\text{cont}} + \mathbf{F}_a^{\text{rep}} + \mathbf{F}_a^{\text{field}}\Big] + \sqrt{2D_T}\, \Delta \mathbf{W}_a(t), \tag{19}$$

where $\gamma_m \in [0, 1)$ is the velocity memory factor, $\mathbf{F}_a^{\text{well}} = -k_w(\mathbf{r}_a - \mathbf{r}_a^{(0)})$ is the harmonic lattice restoring force, $\mathbf{F}_a^{\text{cont}} = -k_c \mathbf{r}_a$ is global radial containment, $\mathbf{F}_a^{\text{rep}} = k_r \sum_{b \neq a} (\mathbf{r}_a - \mathbf{r}_b)/(\|\mathbf{r}_a - \mathbf{r}_b\|^2 + \epsilon_0)^{3/2}$ is softened core repulsion preventing bubble collapse, and $\mathbf{F}_a^{\text{field}} = (c_f k_f + c_s k_s)(\mathbf{r}_a^{\text{target}} - \mathbf{r}_a)$ combines the line-residual drift k_f and centrifugal snap k_s. Simultaneously, each bubble's visual outer shell radius $R_a^{\text{shell}}(t) = R_0 + \alpha_R |Z_a| + \beta_R |\widetilde{g}_a(t)|$ dilates in proportion to local line stress and contracts smoothly onto the static core radius $R_0 + \alpha_R |Z_a|$ as $\mathcal{H}_{\text{ind}}(t) \to 0$.

5.2 The Four Dynamical Regimes and Hybrid Field-Guided Settling

Figure 6a and Table 6 quantify the continuous trajectory of $\mathcal{H}_{\text{ind}}(t)$ and the root-mean-square (RMS) lattice displacement $\Delta r_{\text{rms}}(t) = \big(\frac{1}{n^2} \sum_a \|\mathbf{r}_a(t) - \mathbf{r}_a^{(0)}\|^2\big)^{1/2}$ across the four canonical parameter regimes of `bubbleTrinityApp.js`:

1. **Regime I (*Frozen Lattice*, $k_w = 7.0, k_f = 0$):** Stiff lattice wells lock every bubble at $\mathbf{r}_a^{(0)}$ ($\Delta r_{\text{rms}} = 0.000$), keeping $\mathcal{H}_{\text{ind}}(t) = 1{,}681.89$ completely static.

2. **Regime II (*Slow Drift*, $k_w = 2.6, k_f = 1.0, T = 0.5$):** Moderate field coupling induces small elastic displacements ($\langle \Delta r_{\text{rms}} \rangle = 0.0774$ cell units), smoothly lowering $\mathcal{H}_{\text{ind}}(t)$ by 8.5% to 1,539.22 while preserving crystal order.

3. **Regime III (*Field Storm*, $k_w = 0.8, k_f = 3.2, k_r = 1.0, T = 1.2$):** Strong line-field driving overcomes the local wells ($\max \Delta r_{\text{rms}} = 0.4646$), melting the lattice into a convective fluid that drops $\mathcal{H}_{\text{ind}}(t)$ by 29.0% to 1,194.88.

4. **Regime IV (*Centrifugal Balance*, $k_w = 1.8, k_f = 2.2, k_s = 0.85, T = 0.4$):** Adding targeted snap attraction $k_s = 0.85$ drives displaced bubbles directly toward stress-canceling

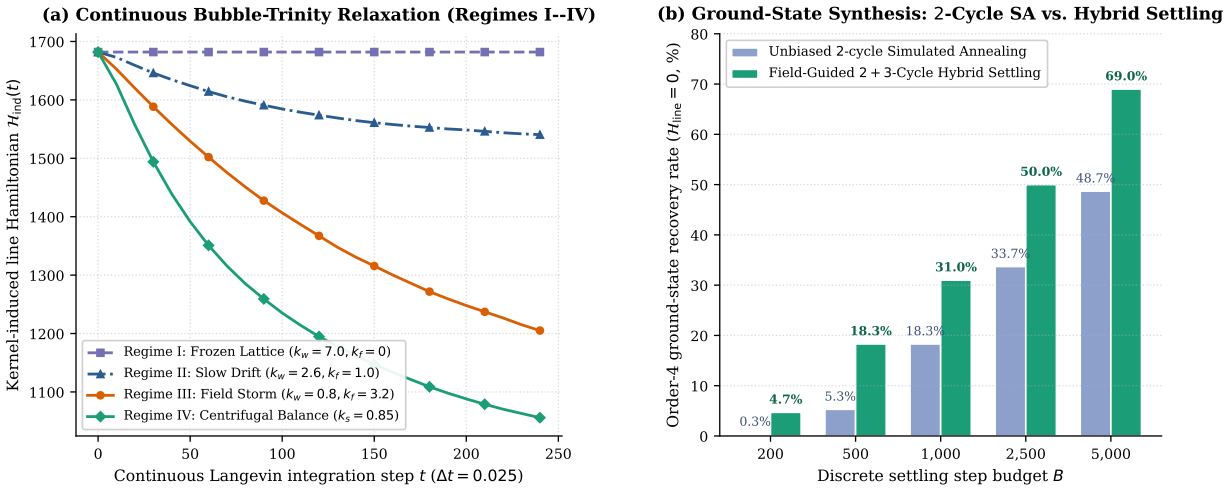


Figure 6: **Continuous Bubble-Trinity Langevin relaxation regimes and hybrid ground-state synthesis benchmarks. (a)** Trajectories of the kernel-induced line Hamiltonian $\mathcal{H}_{\text{ind}}(t)$ across 250 integration steps ($\Delta t = 0.025$) for the four canonical dynamical regimes of Table 6, starting from a 4-swap perturbation of Dürer's 4×4 square ($\mathcal{H}_{\text{ind}}(0) = 1{,}681.89$). **(b)** Order-4 magic ground-state recovery rate ($\mathcal{H}_{\text{line}} = 0$, %, 300 trials per budget) comparing standard unbiased 2-cycle Simulated Annealing against Field-Guided $2+3$-Cycle Hybrid Settling across step budgets $B \in \{200, 500, 1{,}000, 2{,}500, 5{,}000\}$.

Table 6: Parameters and trajectory invariants across the four canonical Bubble-Trinity Langevin regimes at order $n = 4$ (250 steps, $\Delta t = 0.025$), verified by `verify_paper_d.py`.

Dynamical Regime	(k_f, k_r, k_w, k_c)	(γ_m, k_s, T)	$\mathcal{H}_{\text{ind}}(0)$	$\mathcal{H}_{\text{ind}}(250)$	Min $\mathcal{H}_{\text{ind}}$	Mean Δr_{rms}	Max Δr_{rms}
I. Frozen Lattice	$(0.0, 0.0, 7.0, 0.0)$	$(0.94, 0.00, 0.0)$	1681.89	1681.89	1681.89	0.0000	0.0000
II. Slow Drift	$(1.0, 0.4, 2.6, 0.0)$	$(0.82, 0.00, 0.5)$	1681.89	1539.22	1539.22	0.0774	0.1134
III. Field Storm	$(3.2, 1.0, 0.8, 0.4)$	$(0.60, 0.00, 1.2)$	1681.89	1194.88	1194.88	0.2654	0.4646
IV. Centrifugal Bal.	$(2.2, 0.5, 1.8, 0.5)$	$(0.75, 0.85, 0.4)$	1681.89	1050.55	1050.55	0.4281	0.6381

partner cells ($\max \Delta r_{\text{rms}} = 0.6381$), achieving the deepest continuous energy reduction ($\mathcal{H}_{\text{ind}}(t) = 1{,}050.55$, a 37.5% drop).

Finally, to snap the continuous field back onto an exact integer permutation $S \in S_{16}$ with $\mathcal{H}_{\text{line}}(S) = 0$, the Bubble-Trinity *Settle* operator combines two mechanisms inspired by Section 2 and Section 4: (i) *cell-stress prioritization*, which selects the highest-stress cell $a^* = \arg\max_a |g_a(S)|$ with probability 1/2 at each step rather than sampling blind random pairs, and (ii) *multi-cycle mixing*, which proposes a directed 3-cycle with probability 1/4 and a 2-cycle transposition with probability 3/4 under a quadratic annealing schedule $T(t) = T_0(1 - t/B)^2$. As shown in Figure 6b across 300 independent random starts per budget $B \in \{200, 500, 1{,}000, 2{,}500, 5{,}000\}$, Field-Guided $2+3$-Cycle Hybrid Settling consistently outperforms standard unbiased 2-cycle Simulated Annealing, raising the ground-state synthesis rate from 0.3% to 4.7% at $B = 200$, from 5.3% to 18.3% (3.45×) at $B = 500$, from 18.3% to 31.0% at $B = 1{,}000$, from 33.7% to 50.0% at $B = 2{,}500$, and from 48.7% to 69.0% at $B = 5{,}000$.

6 Discussion

Across the four papers of the *Magic Gems* series [11, 10, 9], a unified picture emerges connecting discrete combinatorics, convex geometry, rigid-body mechanics, and statistical-mechanical energy landscapes:

- **From Forward Invariants to Inverse Landscapes:** Whereas Papers I–III showed how the magic line-sum constraints $\mathcal{H}_{\text{line}}(S) = 0$ sculpt the convex hull $K(S)$ (Paper I), diagonalize the inertia tensor via bivariate moment orthogonality (Paper II), and stabilize rigid-body

rotation and tabletop resting equilibria (Paper III), the present paper demonstrates that the operator $L_n = A_n^\top A_n$ governing $\mathcal{H}_{\text{line}}(S) = \mathbf{z}^\top L_n \mathbf{z}$ possesses an equally rich spectral and topological structure away from the ground state.

- **Zero-Mean Spectral Rationalization:** The collapse of the irrational eigenvalues $\lambda_\pm(n)$ of L_n into exact integers $\{0, n-2, n\}$ (even n) or $\{0, n-1, n\}$ (odd n) on $\mathbf{1}^\perp$ (Theorem 2.1) and the universal mean eigenvalue $\bar{\mu}(\widetilde{L}_n) = 2$ (Theorem 2.2) show that the magic-square constraint operator is intrinsically rational on the hyperplane of centred permutations. At the same time, the Low-Mode Insufficiency Proposition (Proposition 2.3) proves that physical moment balance ($\langle X, Z\rangle = \langle Y, Z\rangle = 0$) and diagonal balance ($R_{D_\pm} = 0$) leave a $(2n-3)$-dimensional high-frequency subspace unconstrained.

- **Multiscale Sub-Window Hierarchy:** Within the ground-state manifold $\ker(\widetilde{L}_n)$, our census of sliding-window Shannon entropies $\widetilde{H}_p(k)$ and 2×2 quartet energies (Theorem 3.1 and Proposition 3.2) establishes a second-tier energy hierarchy that separates Dudeney Group I panmagic squares ($W^{\text{int}}_{2\times 2} = W^{\text{tor}}_{2\times 2} = 0$) and Group III associative squares ($B_{\text{comp}} = 0, M_{xyz} = 0$) from generic magic squares, while uncovering the 8 anomalous non-associative octupole-zero squares in Groups IX and X.

- **Topological Frustration and Multi-Cycle Relaxation:** Exhaustive enumeration of S_9 (Theorem 4.2) reveals that even at $n = 3$, the discrete permutation constraint creates 8 strict local minima $S^{(3)}_{\text{trap}}$ at $\mathcal{H}_{\text{line}} = 8$ that are immune to both 2-cycles and 3-cycles and capture 25.47% of all steepest-descent trajectories. At $n = 4$ (S_{16}), where 81.75% of greedy 2-cycle descents become trapped, embedding the discrete lattice into a continuous Langevin stress field and settling with cell-stress-guided $2+3$-cycle moves substantially accelerates ground-state synthesis (Figure 6).

Several natural open problems arise from these findings. First, what is the exact number of spurious 2-cycle and $(2+3)$-cycle local minima in S_{16}, and can every local minimum in S_{16} escape to $\mathcal{H}_{\text{line}} = 0$ via a simultaneous $(2,2)$ double swap as in S_9 (Theorem 4.2)? Second, can the Spectral Rationalization Theorem (Theorem 2.1) be extended to three-dimensional magic cubes S_{n^3} [15, 16] and toroidal panmagic Laplacians? Third, can a continuous Birkhoff-polytope relaxation of $\mathbf{z}^\top L_n \mathbf{z}$ over doubly stochastic permutation matrices be combined with concave entropy penalties to guarantee deterministic convergence to $\ker(\widetilde{L}_n)$ at high orders $n \geq 6$? All interactive simulators, verification scripts, and datasets supporting this work are openly available for further exploration.

Data and Code Availability

The interactive browser laboratory (`physicalSolver`), the exact Python verification suite (`verify_paper_d.py`), the certified JSON census (`results_paper_d.json`), and the figure generation pipeline (`make_figures_d.py`) are publicly available in the consolidated repository https://github.com/kylemath/magicGem and live at https://kylemath.github.io/magicGem/web/physicalSolver/.

Competing Interests

The author declares no competing financial or non-financial interests.

Acknowledgments and Declaration of Generative AI Use

This work grew out of the author's interactive physical and computational explorations of magic-square energy landscapes [12], and forms the fourth part of the *Magic Gems* series alongside [11, 10, 9]. Large language models (OpenAI's ChatGPT, Anthropic's Claude, and Google's Gemini) were used by the author as interactive assistants for symbolic checks, verification scripting,

interactive simulator prototyping, and manuscript copy-editing following George M. Whitesides's iterative outline-and-figure methodology [18]; all mathematical formulations, proofs, and exact computational certificates were verified by the author, who takes full responsibility for the work.

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A Exact Verification Suite and Reproducibility Manifest

All mathematical identities, spectral decompositions, exhaustive S_9 and order-4 censuses, and Langevin trajectories reported in this paper are certified by the standalone Python verification script `verify_paper_d.py`, which outputs the machine-readable JSON dataset `results_paper_d.json`. Specifically:

1. `verify_section2_spectral()` constructs the $(2n+2) \times n^2$ incidence matrix A_n and Laplacian $L_n = A_n^\top A_n$ for all $3 \le n \le 8$, verifies the exact irrational eigenvalues $\lambda_\pm(n)$ on $\mathbb{R}^{n^2}$ (Equation 5), projects L_n onto an orthonormal Householder basis of $\mathbf{1}^\perp$ to verify the exact integer spectrum $\mathrm{spec}(\widetilde{L}_n|_{\mathbf{1}^\perp})$ (Equation 6), certifies $\mathrm{tr}(\widetilde{L}_n) = 2(n^2-1)$ and $\mathbb{E}[\mathcal{H}_{\mathrm{line}}(S)] = 2\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4-1)$ (Theorem 2.2), and checks all residuals and moments of the 4×4 low-mode counterexample S_{low} (Proposition 2.3).

2. `verify_section3_multiscale()` evaluates $\widetilde{H}_p(k)$ ($p \in \{1,2,4\}$, $1 \le k \le n$), interior and toroidal 2×2 quartet energies $W^{\mathrm{int}}_{2\times2}$ and $W^{\mathrm{tor}}_{2\times2}$, complement-pair break counts B_{comp}, and mixed octupole moments M_{xyz} across all 880 Frénicle order-4 magic squares (partitioned by Dudeney Groups I–XII) and across constructive families at $n \in \{3,4,5,6,8\}$, certifying Theorem 3.1 and Proposition 3.2.

3. `verify_section4_swap_landscape()` evaluates $\mathcal{H}_{\mathrm{line}}(S)$ and all 36 transposition deltas $\Delta\mathcal{H}_{\mathrm{line}}(\tau_{ab})$ via Proposition 4.1 across all $9! = 362{,}880$ permutations of S_9, verifies that the 8 Lo Shu states and the 8 antipodal traps $D_4 \cdot S^{(3)}_{\mathrm{trap}}$ are the unique 2-cycle local minima in S_9, confirms that all 168 directed 3-cycles strictly increase $\mathcal{H}_{\mathrm{line}}$ on $S^{(3)}_{\mathrm{trap}}$, computes the exact steepest-descent basin partition (270,442 vs. 92,438, Theorem 4.2), and runs the 2,000-trial steepest-descent and 3-cycle escape benchmark on S_{16}.

4. `verify_section5_bubble_trinity()` integrates the continuous Bubble-Trinity Langevin equations (Equation 19) across Regimes I–IV and benchmarks unbiased 2-cycle Simulated Annealing against Field-Guided 2 + 3-Cycle Hybrid Settling across budgets $B \in \{200, 500, 1{,}000, 2{,}500, 5{,}000\}$.